

JUNJIE LI

Software Engineer

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Languages

Spanish: Professional
Catalan: Professional
English: Professional
Chinese: Native

POLYTECHNIC UNIVERSITY OF CATALONIA
Master of Engineering in Computer Science

09.2024 - 06.2025

UNIVERSITY OF BARCELONA
Bachelor of Engineering in Computer Science

09.2020 - 06.2024

SKILLS

- **Frontend:** JavaScript / TypeScript, React / Next.js, Angular, Vue, Electron, WPF
- **Mobile:** React Native, Kotlin Multiplatform, iOS(Swift), Native SDK
- **Backend:** .Net, Node.js, Nest.js, Laravel, Django, RESTful API, GraphQL
- **Database:** PostgreSQL, MySQL/SQLite, MongoDB, Redis, RabbitMQ, Elasticsearch
- **DevOps:** Git, Docker, K8s, AWS, Azure, Nginx, CI/CD Pipeline, Jenkins, Unit/E2E Testing

Interests

System Architecture Design
Machine Learning, AI & Data Science
High Performance Computing (HPC)

WORK EXPERIENCE

Software Engineer | ALTECH Solutions & Consulting

03.2025 - Present

- Contributed to the design and development of several large-scale public service systems simultaneously.
- supporting urban services, mobility, and infrastructure management.
- Acted as a full-stack engineer projects, working directly with clients to implement end-to-end features.

Research Assistant (Research Collaboration) | BSC & Idaea-CSIC

03.2025 - 05.2026

- Collaborated with researchers from Barcelona Supercomputing Center and Spanish National Research Council to optimize a large-scale hydrological model on HPC clusters using a task-based parallel framework.
- Refactored core simulation components and improved task scheduling for distributed HPC environments.

Software Developer | Herta

04.2024 - 03.2025

- Built AI-powered surveillance software including a .NET WPF desktop application and serverless SaaS services on Azure using .NET and Angular.
- Developed automated testing frameworks and integrated them into CI/CD, improving reliability and deployment efficiency.

Software Developer – iOS & AI (University-Industry Collaboration) | LogMeal Food AI

09.2023 - 07.2024

- Developed a computer vision-based iOS app and SDK for real-time food recognition.
- Built dataset analysis tools and trained a CSWin Transformer model for food classification.

Software Developer | Awaait Artificial Intelligence, S.L.

05.2023 - 04.2024

- Led development of a desktop resource monitoring system using Electron and Node.js.
- Contributed to the development and maintenance of all core company products.

Full Stack Developer | CSI Central de suministros

05.2022 - 10.2022

- Developed, designed and maintained full-stack e-commerce web applications using PHP (Laravel), MySQL and React(Next.js).

PROJECTS

Task-based parallel optimization of hydrological model on HPC

03.2025 - 05.2026

- Optimized hydrological simulations using task-based parallelism, achieving up to 35x performance improvement.
- Research results accepted for presentation at EGU 2026 (European Geosciences Union).

LogMeal food AI

09.2023 - 03.2024

- Collaborated with LogMeal to develop an AI-powered food recognition app and showcased it at MWC 2024.

AI-Driven projects

11.2022 - 02.2023

- Finalist in AiguaDeBarcelona DataChallenge, developing an AI-based water consumption analysis app using PyTorch to forecast trends across industrial, domestic, and tourism sectors.